



ASSIGNMENT No: 19

TITLE: Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.

Problem Statement

Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.

Learning Objectives

To establish database connectivity and retrieve records using JDBC SELECT statements.

Theory

Java Database Connectivity (JDBC) is a standard Java API that enables communication between Java applications and relational databases. It provides a platform-independent mechanism to connect to databases such as MySQL, Oracle, PostgreSQL, and SQL Server. Before executing SQL statements, a JDBC driver is loaded and a connection is established using the DriverManager class. The Connection object represents an active connection to the database, while the Statement object is used to execute SQL queries. The SELECT statement retrieves records from database tables based on specified conditions. The retrieved records are stored in a ResultSet object, which allows sequential access to the data. JDBC simplifies database programming and is widely used in desktop, web, and enterprise applications.

```
import java.sql.*;

public class Main {
    public static void main(String[] args) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/college",
                "root",
                "password");

            Statement stmt = con.createStatement();

            ResultSet rs = stmt.executeQuery("SELECT * FROM student");

            while (rs.next()) {
                System.out.println(
                    rs.getInt("id") + " " +
                    rs.getString("name") + " " +
                    rs.getString("course")
                );
            }

            con.close();
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

```

PS C:\Users\DHARMIT SHAH\Java-Questions> java -cp ".;C:\Users\DHARMIT SHAH\.m2\repository\com\mysql\mysql-connector
.4.0\mysql-connector-j-8.4.0.jar" Exp19.Main
>>
1 Devansh BCA
2 Chirag BCA
3 Dharmit BCA
PS C:\Users\DHARMIT SHAH\Java-Questions>

```

Exercise

1. Create a Java application to display student records from database using SELECT query
2. Create a Java application that connects to a database and displays product details using a SELECT query. Show product ID, product name, quantity, and price.

```

import java.sql.*;

public class Exp1 {
    public static void main(String[] args) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/college",
                "root",
                "root"
            );

            Statement stmt = con.createStatement();

            ResultSet rs = stmt.executeQuery(
                "SELECT * FROM student"
            );

            System.out.println("Student Records:");

            while (rs.next()) {
                System.out.println(
                    "ID: " + rs.getInt("id") +
                    ", Name: " + rs.getString("name") +
                    ", Course: " + rs.getString("course")
                );
            }

            con.close();
        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}

```

```
PS C:\Users\DHARMIT SHAH\Java-Questions> javac Exp19/Exercise/Exp1.java
>> java -cp ".;C:\Users\DHARMIT SHAH\.m2\repository\com\mysql\mysql-connector-j\8.4.0\mysql-connector-j-8.4.0.jar" 9.Exercise.Exp1
>>
Student Records:
ID: 1, Name: Devansh, Course: BCA
ID: 2, Name: Chirag, Course: BCA
ID: 3, Name: Dharmit, Course: BCA
PS C:\Users\DHARMIT SHAH\Java-Questions>
```

```
import java.sql.*;

public class Exp2 {
    public static void main(String[] args) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/shop",
                "root",
                "root"
            );

            Statement stmt = con.createStatement();

            ResultSet rs = stmt.executeQuery(
                "SELECT * FROM product"
            );

            System.out.println("Product Details:");

            while (rs.next()) {
                System.out.println(
                    "Product ID: " + rs.getInt("product_id") +
                    ", Name: " + rs.getString("product_name") +
                    ", Quantity: " + rs.getInt("quantity") +
                    ", Price: " + rs.getDouble("price")
                );
            }

            con.close();
        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Java-Questions> javac Exp19/Exercise/Exp2.java
>>
PS C:\Users\DHARMIT SHAH\Java-Questions> java -cp ".;C:\Users\DHARMIT SHAH\.m2\repository\com\mysql\mysql-connector-j-8.4.0\mysql-connector-j-8.4.0.jar" Exp19.Exercise.Exp2
>>
Product Details:
Product ID: 101, Name: Laptop, Quantity: 5, Price: 55000.0
Product ID: 102, Name: Mouse, Quantity: 20, Price: 700.0
Product ID: 103, Name: Keyboard, Quantity: 10, Price: 1200.0
PS C:\Users\DHARMIT SHAH\Java-Questions> █
```

Conclusion

Thus, the experiment successfully demonstrated the use of JDBC for establishing connectivity between a Java application and a relational database. The program used JDBC components such as DriverManager, Connection, Statement, and ResultSet to establish a connection, execute a SELECT query, retrieve records, and display the required data. This experiment provided an understanding of how Java applications can interact with databases and perform basic data retrieval operations efficiently.